Vishwadhar Kyatham

London, UK | +44 7990 287239 | vishwadharkyatham@gmail.com
LinkedIn | GitHub | Portfolio: vishwadhar0000.github.io

Professional Summary

Applied AI & Machine Learning Engineer with postgraduate training in NLP, Graph Neural Networks, and transformer architectures (BERT). Experienced in engineering end-to-end pipelines from raw corpus preprocessing to asynchronous FastAPI microservices, transductive GCN modeling, and cloud deployments (Render). Targeting AI Engineer and Machine Learning Engineer roles across the UK.

Technical Skills

Languages & Core: Python (NumPy, Pandas, SciPy), SQL, Git, Linux/Bash
Machine Learning & DL: PyTorch, PyTorch Geometric, TensorFlow, Keras, Scikit-learn, GCN, CNN, LSTM
NLP & Transformers: BERT, Hugging Face, TF-IDF, Word2Vec, spaCy, NLTK, Tokenization, Semantic Vector Search
Deployment & MLOps: FastAPI, Uvicorn, REST APIs, Render (IaC), Docker (Basics), Git/GitHub
Evaluation & Metrics: Macro F1-Score, Cross-Validation, Precision/Recall, Cross-Entropy Loss, ROC-AUC, PR-AUC

Featured AI/ML Projects

Graph-Based Fake News Detection System | *PyTorch Geometric, NetworkX, TF-IDF, Scikit-learn*
View Codebase

- Formulated an applied Graph Convolutional Network (GCN) pipeline to evaluate news credibility using inter-document graphs.
- Engineered a dense adjacency matrix via vectorized cosine similarity thresholding (> 0.3) across 500-dimensional TF-IDF vectors.
- Designed a 2-layer `GCNConv` network with dropout ($p = 0.5$), reducing Cross-Entropy loss from 0.4161 to 0.2480.
- Achieved a **0.91 Macro F1-score** and **91.20% test accuracy** across a balanced transductive evaluation set of 1,000 articles.

Production BERT FAQ Conversational Microservice | *Python, BERT, FastAPI, Telegram API, Render*
View Codebase

- Developed an asynchronous NLP microservice using BERT embeddings and FastAPI to automate e-commerce query resolution.
- Built dense vector matching over structured FAQ knowledge bases, enabling contextual query resolution without rigid keyword rules.
- Configured automated cloud deployment pipelines using Render service specifications (`render.yaml`) for continuous uptime.
- Integrated Telegram Bot API webhooks and cosine confidence thresholding to execute graceful fallbacks on low-confidence inputs.

Mobile Click-Fraud Anomaly Detector | *Python, Scikit-learn, Feature Engineering* View Codebase

- Engineered a client-side anomaly detection model to detect bot clicks and touch injection attacks on mobile telemetry feeds.
- Extracted behavioral features (touch coordinate variance, sequence intervals) to capture non-human interaction patterns.
- Benchmarked lightweight tree-based classifiers to support offline, battery-efficient inference without remote network round-trips.

Education

University of East London, London, UK 2024 – 2026
Master of Science in Artificial Intelligence (Merit / 2:1 Standing: 67%)
Relevant Modules: Machine Learning, Deep Learning, Natural Language Processing, AI Ethics, Research Methods

Narsimha Reddy Engineering College, India 2018 – 2022
Bachelor of Technology in Computer Science & Engineering (CGPA: 6.5 / 10.0)

Certifications

Generative AI & LLM Foundations – LinkedIn Learning
Mathematical Foundations of Machine Learning – Udemy
Natural Language Processing – Great Learning
Introduction to AI – Great Learning